

ABHISHEK SHUKLA

Senior Data Engineer · AI Engineer

AWS · PySpark · Apache Iceberg · Redshift · Rust · Amazon Bedrock · RAG · Text-to-SQL · AI Agents · LLM Evals
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SUMMARY

Data engineer with **5+ years** on AWS, building the data platforms AI depends on and the LLM systems that run on them. Currently **Associate Tech Architect** on a **~1 TB/day** serverless telecom analytics platform delivered with AWS ProServe, and shipping **RAG, text-to-SQL and agent systems on Amazon Bedrock** with eval-gated releases. Grew from **intern to architect** at Quantiphi.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Rust, TypeScript
Data Engineering: Apache Spark, Apache Iceberg, EMR Serverless, Parquet, ETL/ELT, CDC, streaming, star schema and SCD Type 2 modeling, data quality, late-arriving data and backfills
AWS: S3, Lambda, Glue, EMR, Redshift, Athena, DynamoDB, Aurora PostgreSQL, Kinesis, DMS, SQS, SNS, EventBridge, Step Functions, Lake Formation, CloudWatch, KMS
AI and LLM: Amazon Bedrock, Anthropic Claude, RAG (chunking, embeddings, hybrid retrieval, reranking), text-to-SQL, vector stores (OpenSearch, pgvector, S3 Vectors), prompt engineering
Agents and Evals: Bedrock AgentCore, Model Context Protocol (MCP), tool calling, OAuth, LLM-as-judge eval harnesses, CI regression gating, OpenTelemetry
Databases: Redshift, Aurora PostgreSQL, ClickHouse, DynamoDB, MySQL
DevOps and Orchestration: Terraform, Apache Airflow, Step Functions, GitLab CI/CD, GitHub Actions, Docker, pytest, SonarQube

CERTIFICATIONS

AWS Certified Solutions Architect - Professional (2026) · [Verify on Credly](#)
AWS Certified Data Engineer - Associate (2026) · [Verify on Credly](#)
Snowflake SnowPro Core (2023)

EXPERIENCE

Quantiphi Analytics Solutions, Mumbai	Feb 2021 - Present
Associate Tech Architect, Data & AI Engineering	Jan 2026 - Present
<i>RAN performance analytics for a UK telecom operator (with AWS ProServe), plus AI systems over governed data.</i>	
Telecom analytics platform: Own the enrichment, schema management and aggregation stages of a serverless platform processing ~1 TB/day of RAN counter data from 19,500 mobile sites, serving 150,000+ cell metrics across 6 time granularities from Apache Iceberg. Multi-vendor schema registry: Built pluggable adapters for 3 vendors and 5 stream formats (Ericsson, Nokia, Huawei) feeding a versioned DynamoDB registry that flags breaking changes between vendor releases and propagates schema to Iceberg, ClickHouse, Redshift and Aurora via Step Functions. New counters ship with no code deploy. Rust rewrite, 16x faster: Rewrote a Python ETL Lambda in Rust, cutting runtime from 8 minutes to 30 seconds and moving it well clear of the 15-minute Lambda limit. Event-driven enrichment: Batchier Lambda handling 100K+ SQS messages per invocation (parallel poll, FIFO send, delete workers), cross-file deduplication in DynamoDB, and Iceberg compaction that cut file count 10x, lowering query planning time and S3 LIST cost. Performance debugging: Traced a 100% pre-prod failure rate in the aggregation Lambda to unbounded DynamoDB existence checks. Early-exit lookups and parallelized calls saved 85-230 s per invocation with zero logic changes and 531 tests green. Added self-healing re-aggregation for data arriving up to 90 days late. RAG and text-to-SQL: Lead natural-language querying over governed S3, Iceberg and Redshift data with cited answers on Amazon Bedrock (Claude), from PoC to production. Built the eval harness (golden datasets, LLM-as-judge, retrieval and faithfulness metrics) that gates every release in CI. Agent on Bedrock AgentCore: Shipped an internal agent in use by internal teams, with tools served through Gateway (including a custom MCP server), per-actor memory with tested tenant isolation, OAuth/JWT authentication with policy-based authorization on sensitive tools, and end-to-end OpenTelemetry tracing.	
Senior Data Engineer	Apr 2023 - Dec 2025
Multi-tenant lakehouse: Architected Apache Iceberg on S3 with Glue Data Catalog, Lake Formation governance and an Aurora serving layer for 15+ tenants, with tenant isolation, schema evolution and backfills built in. Audience counting at scale: Designed a Roaring Bitmap system on EMR and Lambda over billions of rows of affinity data, returning sub-second cardinality queries where standard aggregation timed out. Near-real-time ingestion: Built Kinesis Firehose, DMS and Redshift pipelines that cut reporting latency from 6 hours to under 30 minutes, with late-data handling and idempotent reprocessing. Warehouse performance: Tuned Redshift sort/distribution keys and materialized views, taking complex BI queries from 3 minutes to under 30 seconds and lowering cost per query. Unified two marketplaces into one star schema with conformed dimensions and SCD attributes. GenAI reporting app: Shipped a Bedrock application that writes narrative report insights, turning a 2-day manual process into under 2 hours. Owned it from architecture to production. API integrations: Integrated 5+ third-party APIs through Glue and Lambda for 100% daily reporting coverage.	

Data Engineer

Aug 2021 - Mar 2023

- **Sports analytics ETL:** Designed a PySpark pipeline (Glue, Step Functions) processing **50+ GB/day** of computer-vision data for a **Bundesliga** client, and built **5 QuickSight dashboards** adopted by coaching staff for match analysis.
- **Contact-center analytics:** Built a post-call analytics pipeline that lifted IVR call containment **12% across 15+ KPIs**.

Data Engineer Intern

Feb 2021 - Aug 2021

- Prototyped a Spark and AWS pipeline that cut manual computer-vision annotation time by **35%**.

PROJECTS

Governed LLM Analytics · github.com/abhis5/governed-llm-analytics

Text-to-SQL where the LLM emits **validated query intent over a governed vocabulary** instead of raw SQL, so it cannot run a query the governance layer has not approved. Includes an eval harness (golden datasets, LLM-as-judge on Bedrock Claude) wired into CI, plus a decision log of rejected designs.

Paisa · React personal finance app with India-specific features: UPI categorization, financial-year handling and bank CSV import.

EDUCATION

Master of Computer Applications (MCA), Vivekanand Education Society's Institute of Technology, Mumbai (2021)

Bachelor of Computer Applications (BCA), Virendra Swarup Institute of Computer Studies, Kanpur (2018)